Analytics & Big Data

Session 6: Machine learning I

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🎯 **Learning objectives**

- **Distinguish** between supervised and unsupervised machine learning approaches and explain the generalization problem in supervised machine learning.
- **Describe** the workflow of supervised machine learning, including feature engineering, train–test splitting, model training, cross-validation, and evaluation.
- **Connect** conceptual machine learning procedures to Python implementations, including preprocessing, model training, and evaluation using scikit-learn. (*see exercise*)

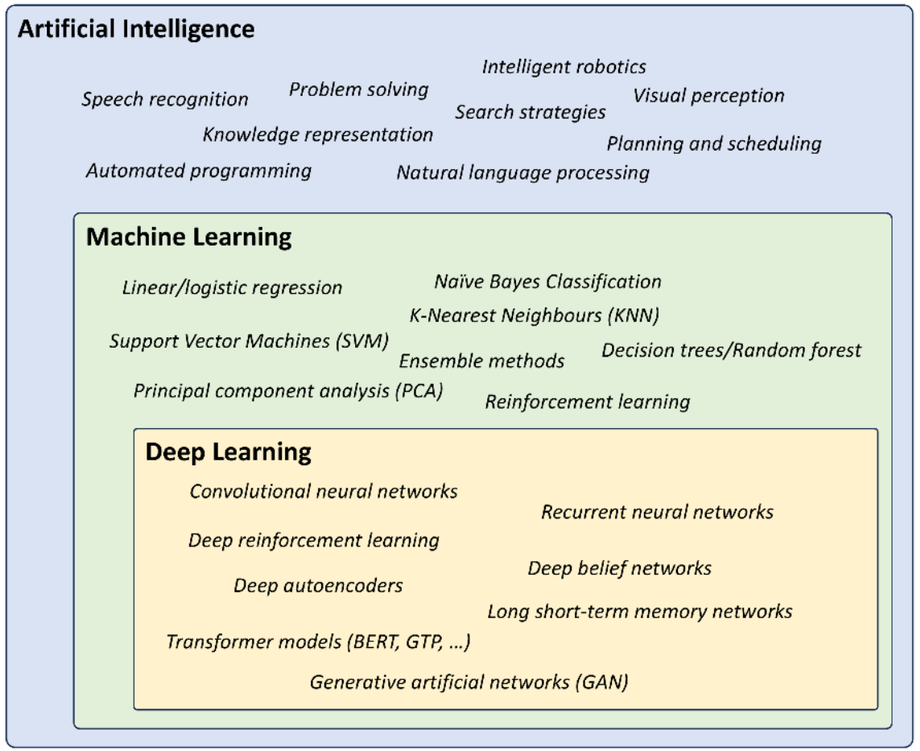
Foundations

Distinguishing concepts

Artificial Intelligence (AI) involves techniques that equip computers to emulate human behavior, enabling them to learn, make decisions, recognize patterns, and solve complex problems in a manner akin to human intelligence.

Machine Learning (ML) is a subset of AI, uses advanced algorithms to detect patterns in large data sets, allowing machines to learn and adapt. ML algorithms use supervised or unsupervised learning methods.

Deep Learning (DL) is a subset of ML which uses neural networks for in-depth data processing and analytical tasks. DL leverages multiple layers of artificial neural networks to extract high-level features from raw input data, simulating the way human brains perceive and understand the world.



Generative AI (genAI) is a subset of DL models that generates content like text, images, or code based on provided input. Trained on vast data sets, these models detect patterns and create outputs without explicit instruction, using a mix of supervised and unsupervised learning.

Supervised and unsupervised learning

Supervised Learning:

- **Goal: predict a target or outcome variable**
- **Training data where target value is known**
- **Score to test data where value is not known**

DEP_TIME	DEST	DISTANCE	FL_DATE	ORIGIN	Weather	DAY_WEEK	DAY_OF_MONTH	FlightStatus
1455	JFK	184	01.01.2004	BWI	0	4	1	ontime
1640	JFK	213	01.01.2004	DCA	0	4	1	ontime
1245	LGA	229	01.01.2004	IAD	0	4	1	ontime
1709	LGA	229	01.01.2004	IAD	0	4	1	ontime
1035	LGA	229	01.01.2004	IAD	0	4	1	ontime
839	JFK	228	01.01.2004	IAD	0	4	1	ontime
1243	JFK	228	01.01.2004	IAD	0	4	1	ontime
1644	JFK	228	01.01.2004	IAD	0	4	1	ontime
1710	JFK	228	01.01.2004	IAD	0	4	1	ontime
2129	JFK	228	01.01.2004	IAD	0	4	1	ontime
2114	LGA	229	01.01.2004	IAD	0	4	1	ontime
1458	JFK	213	01.01.2004	DCA	0	4	1	ontime
932	LGA	214	01.01.2004	DCA	0	4	1	ontime
1228	LGA	214	01.01.2004	DCA	0	4	1	ontime

HousePrice	HouseSize	HouseAge	LotSize	RiverSide	CrimeRate	Industrial
244	6.064	59.1	0	1	.13587	10.59
212	6.176	72.5	0	0	.13158	10.01
222	6.358	52.9	30	0	.1029	4.93
134	6.103	85.1	0	0	705.042	18.1
238	5.877	79.2	0	0	180.028	19.58
174	5.594	36.8	12.5	0	.13554	6.07
222	6.316	38.1	0	0	.05083	5.19
262	6.718	17.5	22	0	.19073	5.86
140	6.174	93.6	0	0	.2909	21.89
190	5.985	45.4	0	0	.05497	5.19
178	7.393	99.3	0	0	824.809	18.1

Unsupervised Learning:

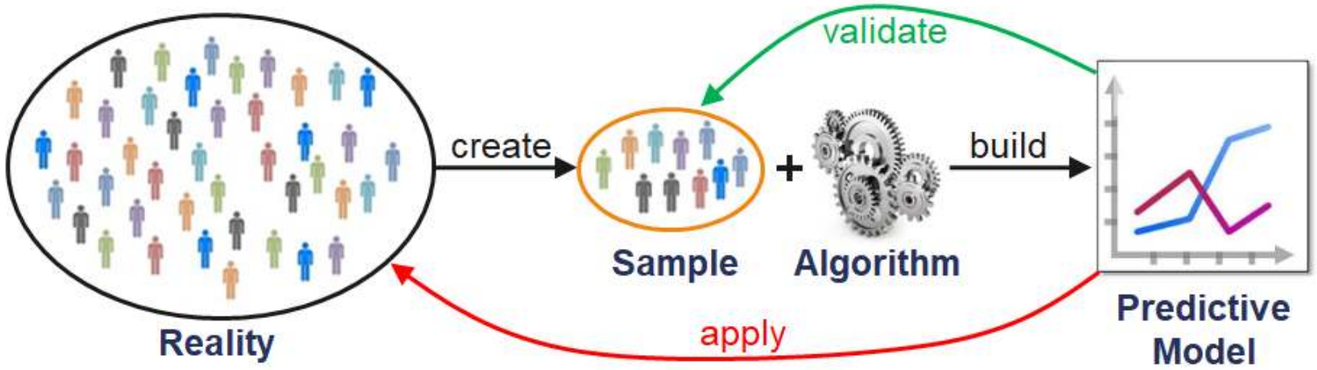
- **Goal: detect patterns**
- **Grouping of cases based on similarities in input values**
- **There is no target (outcome) variable**

Note: Our focus will be primarily on supervised machine learning:

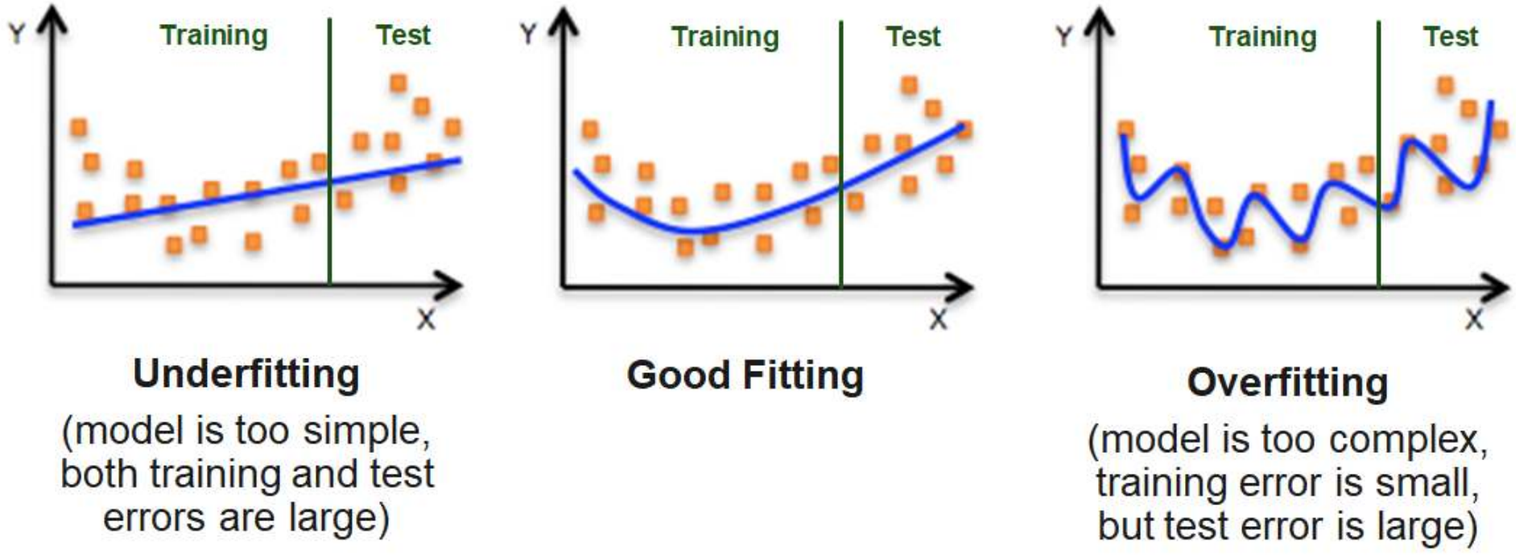
$$\underbrace{\text{Dataset}}_{\text{Features, Targets}} + \underbrace{\text{Learning Algorithm}}_{\text{Model Class + Objective + Optimizer}} \rightarrow \text{Predictive Model}$$

The generalization problem

The traditional analytics process

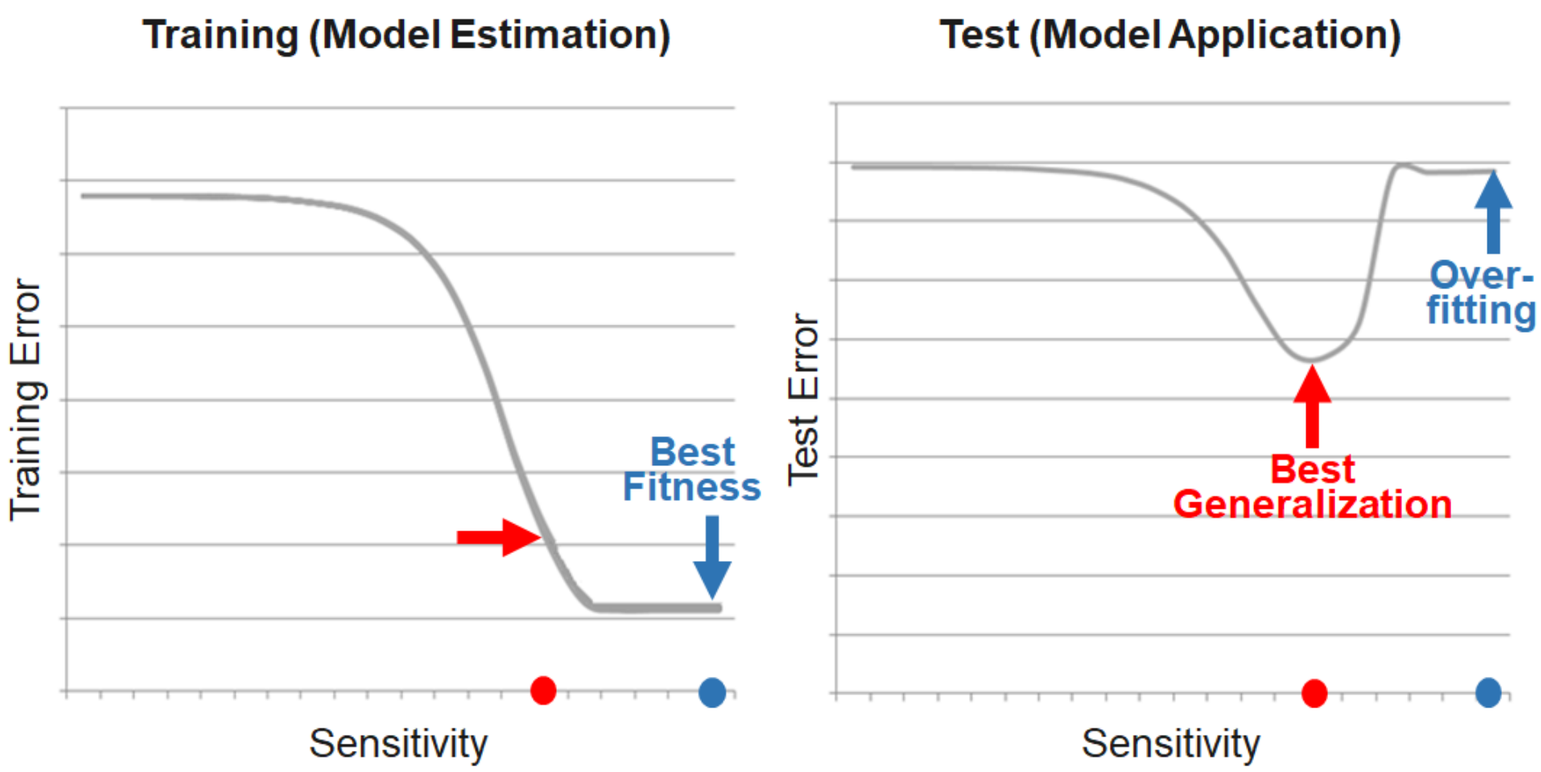


Over- and underfitting



Due to the problem of overfitting, the main goal is to maximize the prediction quality and not to fit the data that is used for the model estimation as well as possible. This is equivalent to minimizing the risk that the model will have weak predictive ability.

Best fit vs. best generalization

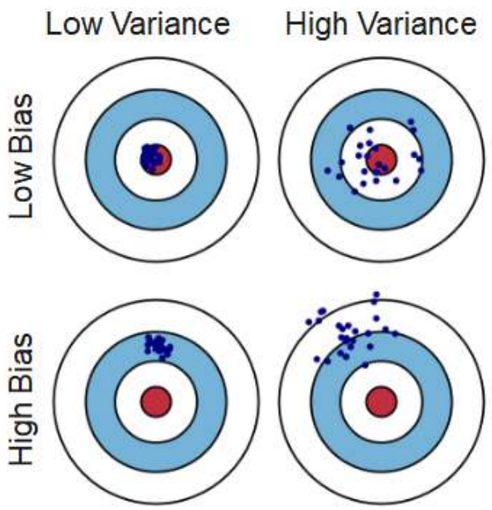


The bias-variance tradeoff

The prediction error is influenced by three components:

> Error = Bias + Variance + Noise

- Bias is the inability of the used method to learn the relevant relations between the inputs and the outputs. It reflects the method quality, e.g. if a method only produces linear models.
- Variance represents the deviation resulting from the sensitivity of the created model to small fluctuations in the data.
- Typically, there is a tradeoff between bias and variance.
- Noise is everything that arises from random variations in the data. It cannot be controlled.



Generalization problem

Generalization problem (concise): Modern machine learning models are highly flexible and trained on large datasets. While this enables them to fit complex patterns, it also creates a risk:

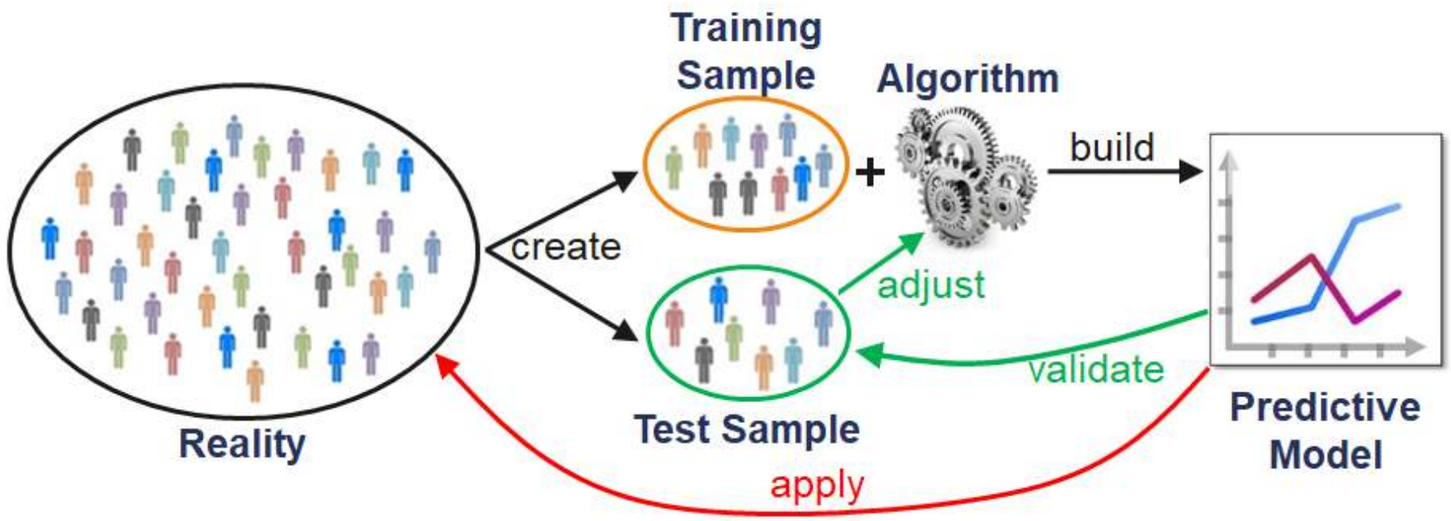
> the model may **fit the training data extremely well** but fail to capture patterns that hold more generally.

The core question is therefore:

> How can we tell whether a model has learned something **generalizable**, rather than just memorizing the data?

The machine learning workflow

The supervised machine learning workflow



Partitioning the data

Data is partitioned into **training** and **test sets** to assess whether a model **generalizes** beyond the data used for estimation.

- The model is **trained** on the training data
- Its performance is **evaluated** on the test data

If performance is substantially worse on the test data, the model is likely **overfitting**—capturing patterns specific to the training data rather than general relationships.

How can the data be split?

- **Random / stratified sampling** *(common, but may limit reproducibility)*
- **Predefined lists**
- **Rule-based splits** *(e.g., first/last observations, time-based rules)*

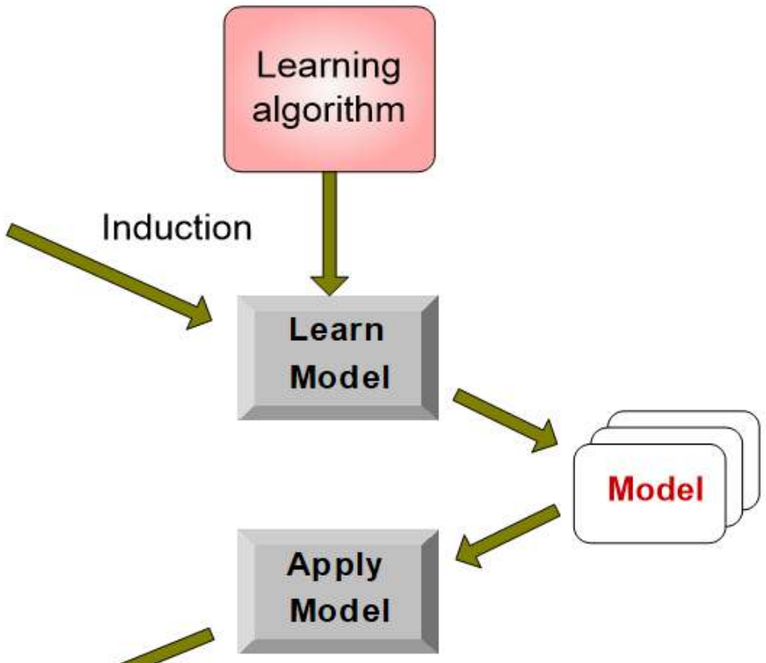
Applying training and test data

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set

Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set



— Source: http://www.cs.kent.edu/~jin/BigData/Lecture10-ML-Classification.pptx

Problems with fixed training and test samples

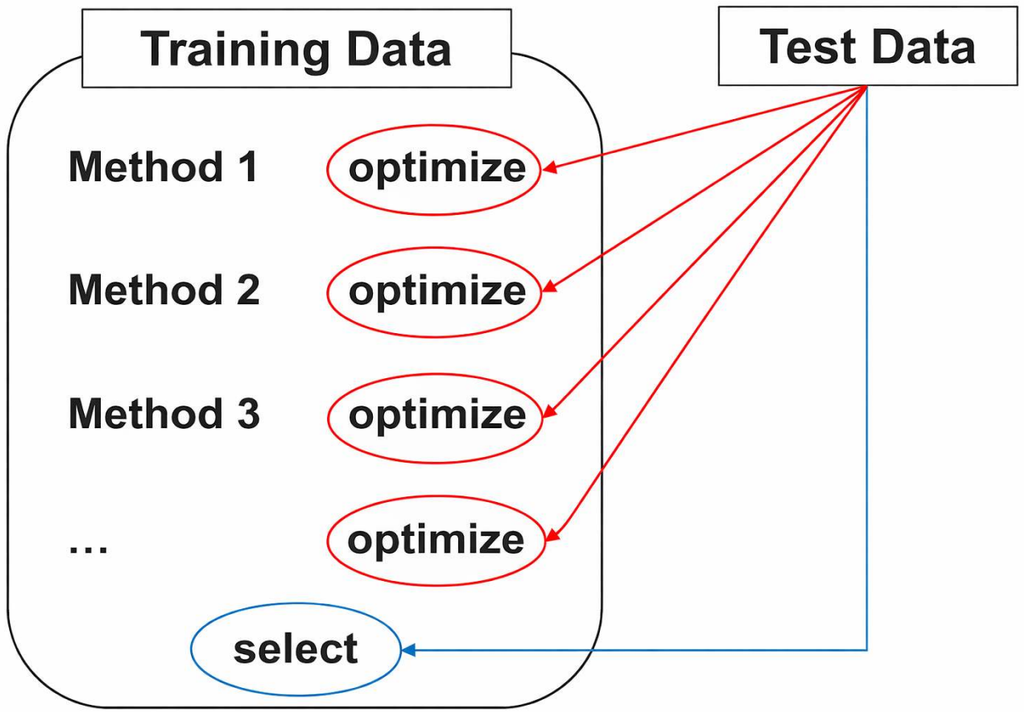
Test data is used for two things:

1. Optimize the model training
2. Select the best model via testing the model quality

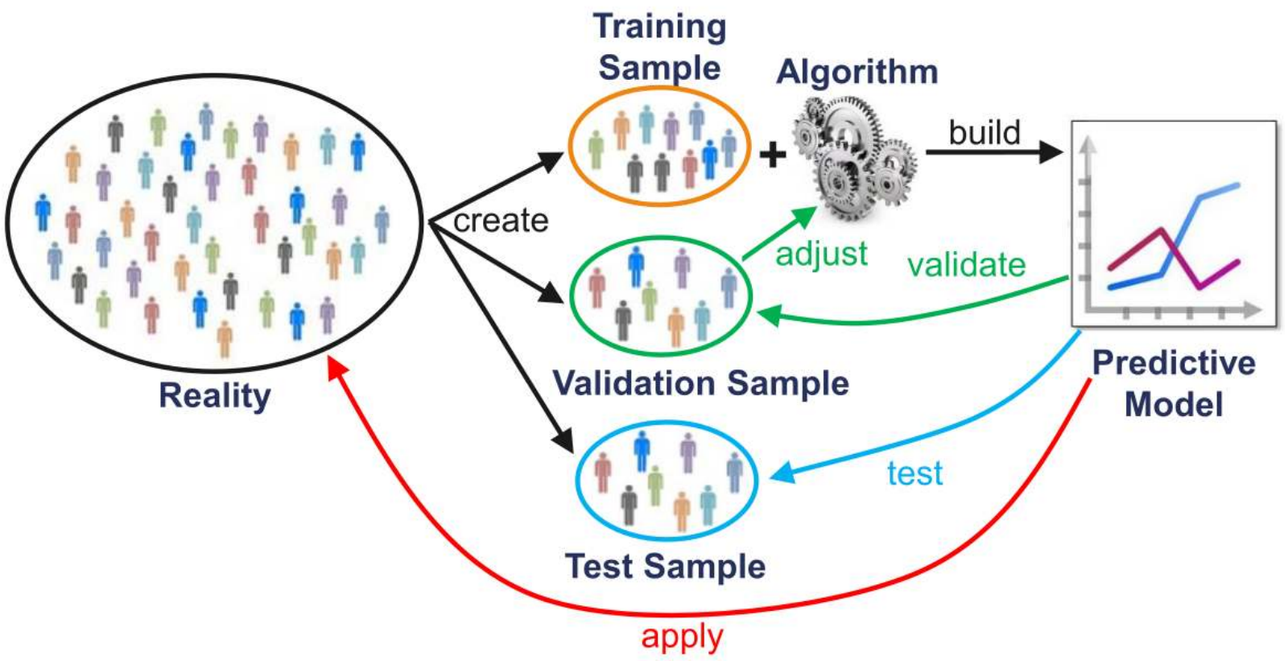
This contradicts the idea of independent testing and results in:

- Endogenization of the test data
- Selection Bias

Rule: NEVER use any information from the test data for model training!



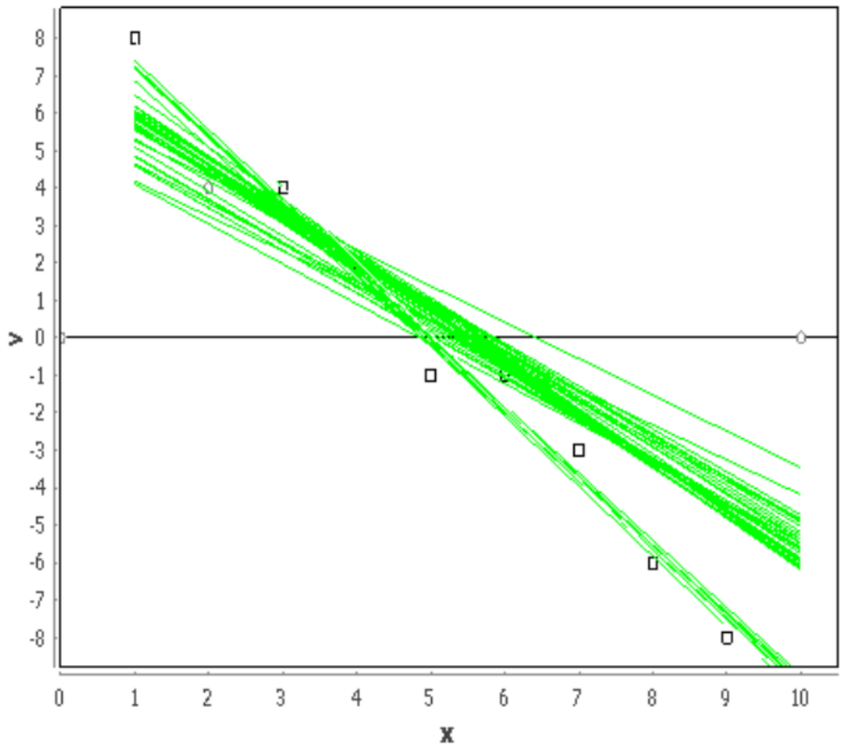
Addressing the endogeneity problem



Selection bias

Training and test error can be highly variable, depending on precisely which observations are included in the training set and which observations are included in the validation set (**Selection Bias**).

Example of different OLS models as a result of different samples:

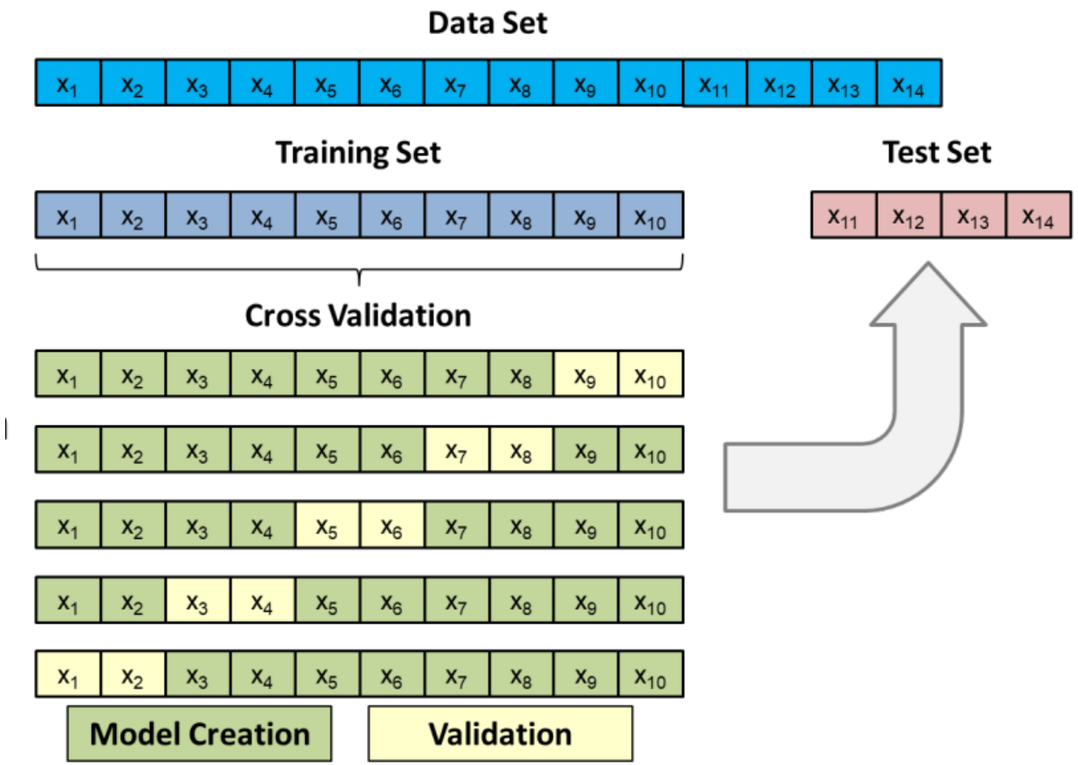


To avoid such problems, one can use so-called resampling methods.

Cross validation

Cross validation can be used for model selection and adjustment. In these cases, cross validation is applied to the training data set. For every iteration, k-1 folds are used for model fitting and the remaining fold for testing the model (Validation). Every time, the quality measure (e.g. accuracy) for the validation fold is captured. At the end of this step, the average and the standard deviation of the measures are calculated. The best model is the one with the best ratio in high average and low standard deviation.

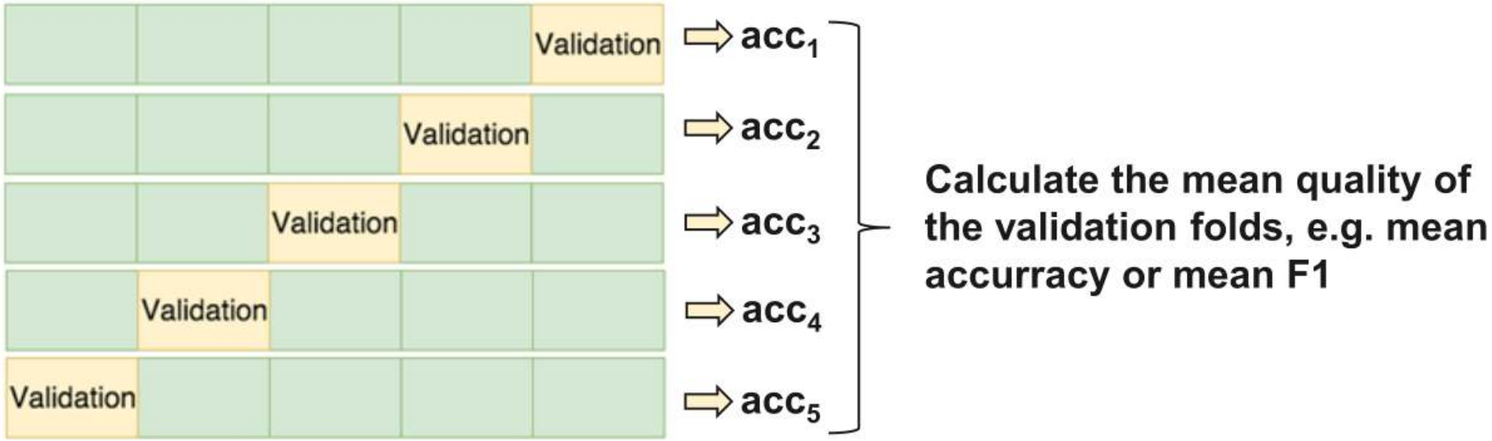
Once the model type and its optimal parameters have been selected, a final model is trained using these hyper-parameters on the full training set, and the generalization quality is measured on the test set.



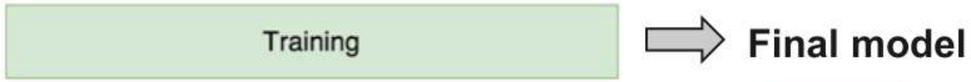
Cross validation and grid search

1. **Partition the Dataset into a training and test set**

2. **For every hyperparameter value combination apply cross validation**



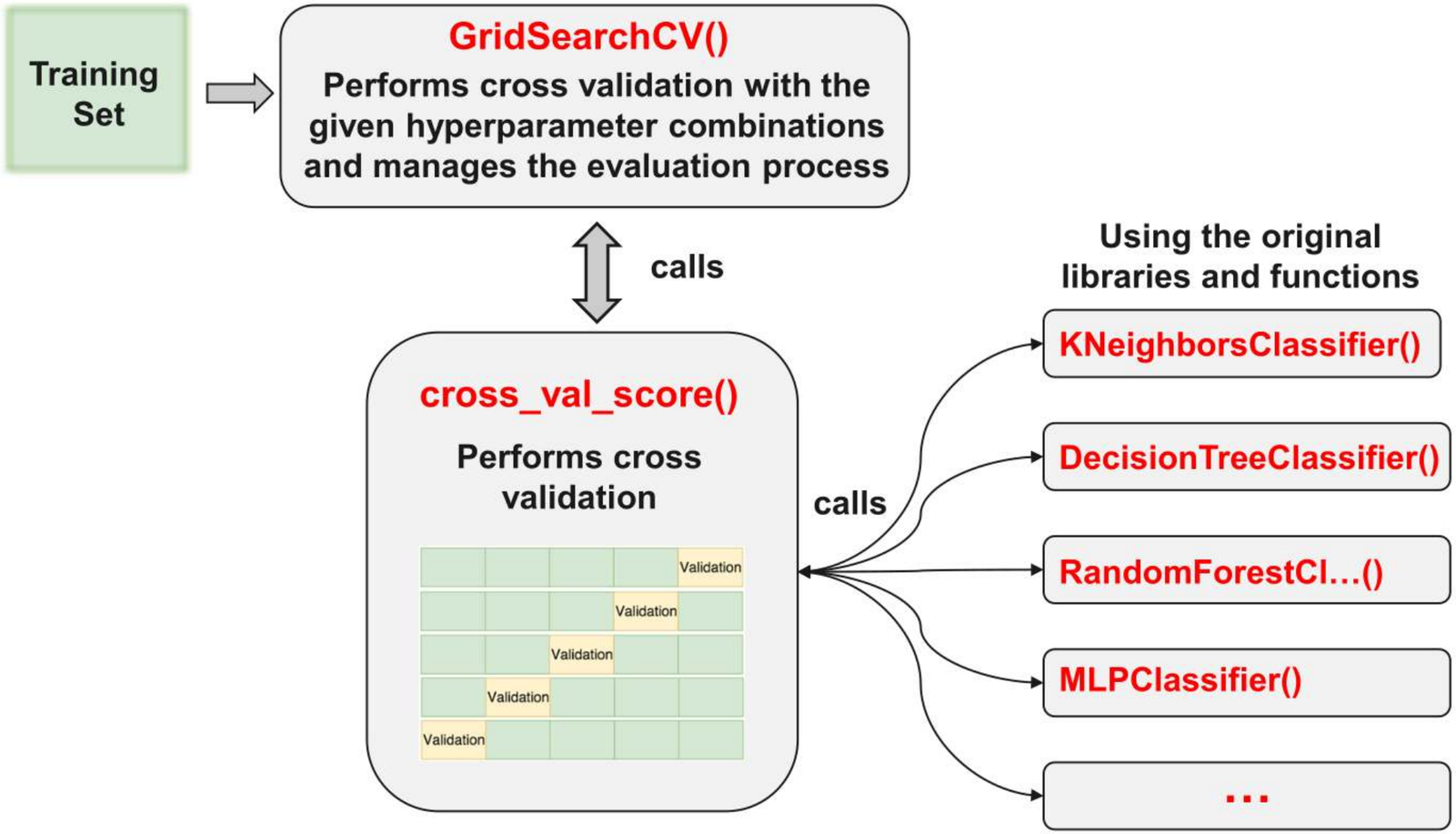
3. **For the combination with the highest (mean) quality calculate the final model with the complete training set**



4. **Test the final model with the test set** Testing

5. **Compare the accurracies of training and test with regard to overfitting**

Cross validation and grid search in Python



Feature leakage

Feature leakage occurs when a model uses **information that would not be available at prediction time**.

- The model appears to perform **very well on training/test data**
- But fails in real-world application
- Typical causes:
  - using **future information** (e.g., outcome-related variables)
  - leakage from **test set into training**
  - features that are **proxies of the target**

Example

- Predicting customer churn
- Feature: *"account closed date"*

→ This directly reveals the outcome

> **Key takeaway**: Only use features that are **available at the time of prediction**

Feature engineering

The feature engineering process

Feature engineering is the process of using domain knowledge of the data to create features that make machine learning algorithms work. If feature engineering is done correctly, it increases the predictive power of machine learning algorithms by creating features from raw data that help facilitate the machine learning process.

A feature (variable, attribute) is depicted by a column in a dataset. Considering a generic two-dimensional dataset, each observation is depicted by a row and each feature by a column, which will have a specific value for an observation:

Columns (Features)

Rows (Observations)

	Feature 1	Feature 2	Feature 3
0	18.38	58.06	-44.21
1	16.10	35.39	77.35
2	23.14	230.44	-71.79
3	24.77	206.83	46.13

Features can be of two major types.

- **Raw features** are obtained directly from the dataset with no extra data manipulation or engineering.
- **Derived features** are usually obtained from feature engineering, where we extract features from existing data attributes. A simple example would be creating a new feature “Age” from an employee dataset containing “Birthdate”.

— Source: Sarkar, D.: Understanding Feature Engineering, towardsdatascience.com and Shekhar, A.: What Is Feature Engineering for Machine Learning?, medium.com.

Variants of feature engineering

1. Transformation

- convert features (e.g., birth date → age)
- build lag structures (e.g., time-lags)
- normalization / standardization / scaling

2. Type Conversion

- if numerical type is needed, transform categorical into numerical data using dummy features
- if categorical type is needed or more informative, discretize numerical features (e.g., income → poor / rich classes)

3. Feature Combination

- create interaction features (e.g., school_score = num_schools × median_school with num_schools = number of schools within 5 miles of a property and median_school = median quality score of those schools)
- combine categories (e.g., when there are very few observations or too many dummy features)

4. Feature Composition

- build ratios (e.g., returns from prices)
- Principal Component Analysis (Dimensionality Reduction)

Scaling

Most datasets contain features highly varying in magnitudes, units and range.

Most machine learning algorithms have problems with this because they use distance measures or calculate gradients. The features with high magnitudes will weigh in a lot more in the distance calculations than features with low magnitudes and gradients may end up taking a long time or are not accurately calculable.

To overcome this effect, we scale the features to bring them to the same level of magnitudes. The two most discussed scaling methods are Normalization and Standardization.

Normalization (Min–Max Scaling: values in [0,1])

$$\hat{X}_i = \frac{X_i - X_{\min}}{X_{\max} - X_{\min}}$$

Standardization (Z-score Scaling: values with mean 0 and stdandard deviation 1)

$$\hat{X}_i = \frac{X_i - \mu}{\sigma}$$

For interpretation, inverse transformations are possible: $X_i = \hat{X}_i \cdot (X_{\max} - X_{\min}) + X_{\min}$ and $X_i = \hat{X}_i \cdot \sigma + \mu$, respectively.

Type vonversion (encoding)

Many machine learning algorithms cannot work with categorical data directly. To convert categorical data to numbers, there exist two variants:

Label encoding refers to transforming the word labels into numerical form so that the algorithms can understand how to operate on them. Every categorical value is assigned to one numerical value, e.g. young → 1, middle_age → 2, old → 3. This only works in specific situations where you have somewhat continuous-like data, e.g. if the categorical feature is ordinal.

One hot encoding is a representation of a categorical variable as binary vectors. Every categorical value is assigned to an artificial binary variable. If the corresponding categorical value occurs in a data row the value of its binary replacement is equal to 1 else 0, e.g.

Country
Japan
U.S
India
China
U.S
India

Country_Japan	Country_U.S	Country_India	Country_China
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1
0	1	0	0
0	0	1	0

It is usual when creating dummy variables to have one less variable than the number of categories present to avoid perfect collinearity (dummy variable trap).

Example of feature engineering (I)

Data sets often contain date/time features. These features are rarely useful in their original form because they only contain ongoing values. However, they can be useful for extracting cyclical factors, such as weekly or seasonal effects. Suppose, we are given a data “flight date time vs status”. Then, given the date-time data, we have to predict the status of the flight.

But the status of the flight may depend on the hour of the day, not on the date-time. To analyze this, we will create the new feature “Hour_Of_Day”. Using the “Hour_Of_Day” feature, the machine will learn better as this feature is directly related to the status of the flight.

	Date_Time_Combined	Status
0	2018-02-14 20:40	Delayed
1	2018-02-15 10:30	On Time
2	2018-02-14 07:40	On Time
3	2018-02-15 18:10	Delayed
4	2018-02-14 10:20	On Time

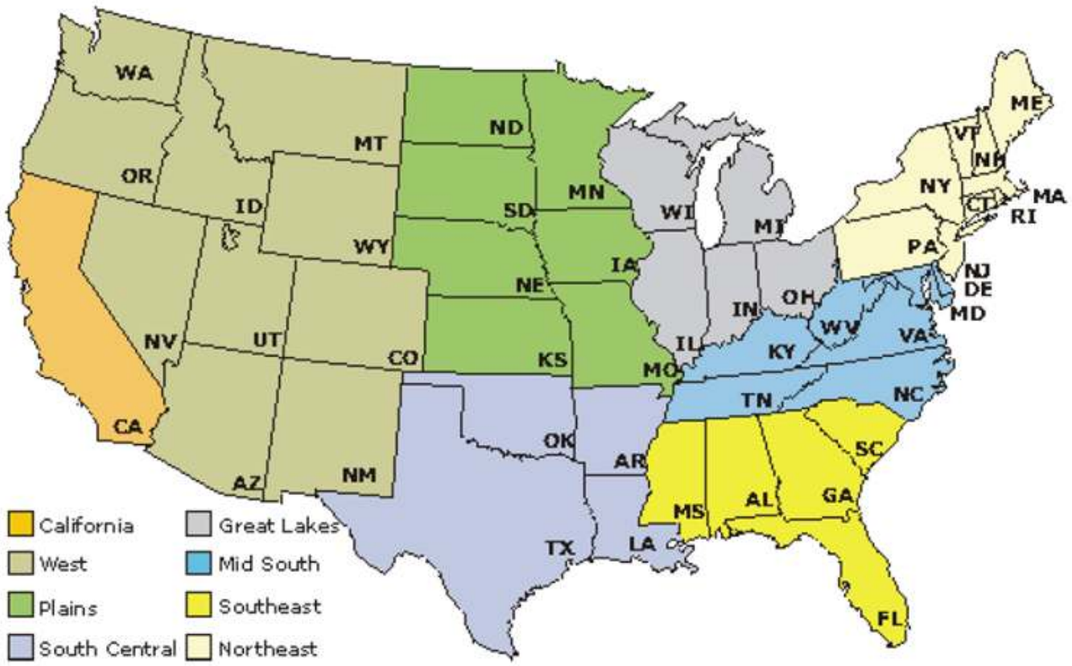
	Hour_Of_Day	Status
0	20	Delayed
1	10	On Time
2	7	On Time
3	18	Delayed
4	10	On Time

— Source: Shekhar, A.: What Is Feature Engineering for Machine Learning?, medium.com.

Example of feature engineering (II)

Suppose we are given the latitude, longitude and other data with the objective to predict the target feature “Price_Of_House”. Latitude and longitude are not of use in this context if they are alone. So, we will combine the latitude and the longitude to make one feature.

In other cases, it might be appropriate to transform latitude and longitude into categories which reflect regions, for example.



Example of feature engineering (III)

Suppose we are given a feature “Marital_Status” and other data with the objective to classify customers into “Creditworthy” and “Not_Creditworthy”. In the data set the martial status has many different values, for example:

- single living alone
- single living with his parents
- married living together
- married living separately
- divorced
- divorced but living together
- registered partnerships
- living in marriage-like community
- widowed
- …

To avoid a transformation into too many and maybe dominating dummy features, we can group the similar classes, e.g. in single, married, widowed.

If there exist some remaining sparse classes which cannot be assigned in a meaningful way they can be joined into a single “other” class.

Summary

- The focus is on **supervised machine learning**, a particular form of machine learning within **AI/ML/DL**, where models learn relationships between **features (X)** and **targets (y)** from labeled data. In contrast, **unsupervised learning** identifies structure in data without predefined labels.
- Modern machine learning models are often **highly flexible** and trained on **large datasets**, creating the risk of **overfitting**. The central challenge is the **generalization problem**: distinguishing true patterns from those specific to the training data.
- The **machine learning workflow** addresses this challenge by separating model development and evaluation: feature engineering, data splitting (training vs. test), model training, and performance assessment.
- Reliable model evaluation requires strict separation of training and test data, supported by techniques such as **cross-validation** and **hyperparameter tuning**, while avoiding issues like **selection bias** and **feature leakage**.
- **Feature engineering** is a critical step that transforms raw data into meaningful representations (e.g., scaling, encoding, feature construction), often having a stronger impact on performance than the choice of model.

Survey: Session 6

https://forms.gle/AFmpWcopjMtGfNif6

Note: Responses may be analyzed and published in anonymized form.

Please complete the survey before you leave today — thank you 🙏